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Integrating mobility, travel survey, and malaria case data to understand drivers of malaria importation to Zanzibar, 2022–2023

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Abstract

Background Zanzibar has achieved historic reductions in malaria incidence, but high connectivity to mainland Tanzania and imported cases remain a challenge to "last mile" malaria elimination.

Methods To understand factors driving malaria importation, prospective surveys were administered to patients at 94 health facilities across Zanzibar's main island, Unguja, from May 2022 to December 2023, and to travellers at the Zanzibar-bound ferry terminal in Dar es Salaam from August to December 2024. In addition, digital mobility data—Meta colocation and movement distribution—were analysed to examine movement patterns between Unguja and mainland Tanzania. These data were integrated with rainfall data to explore the seasonality of human movement and travel-associated malaria risk.

Results Of 3,875 total malaria cases presenting to health facilities in Unguja, 1,172 reported travel to mainland Tanzania within the past 30 days (283/511 (55%) in 2022, 889/3,348 (27%) in 2023). Among these, travel to Tanga (20%), Dar es Salaam (20%), and Morogoro (15%) were most common. Children and females had higher odds of having malaria associated with travel to the mainland (aOR 3.2, 95% CI: 2.2–4.8 and aOR 1.5, 95% CI: 1.3–1.8, respectively, compared to those without mainland travel), while nightwatchmen and students with malaria were more likely to report travel within Zanzibar. Twenty-eight percent of travel-associated malaria cases reported primary residence on the mainland. However, Zanzibari residents who reported travel to mainland regions with high or moderate malaria risk during the dry season made up the highest proportion of travel-related malaria cases (n = 309, 32%). Meta movement distribution data shows that long-distance travel off Unguja decreases during the rainy seasons. In addition, imported cases correlated with rainfall at their mainland destination, rather than rainfall in Zanzibar. With different biases, data from both Meta colocation and ferry surveys approximated the proportional makeup of at-risk travel from different geographies relatively well. In particular, Meta colocation data highlighted travel between Tanga and Unguja, likely via private boats, not captured in the ferry data.

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Conclusions Movement flows and seasonal rainfall patterns drive imported malaria in predictable ways that can be harnessed to target high-risk travellers for intervention.

Keywords Malaria importation, Zanzibar, Tanzania, Human migration, Malaria elimination

Background

Zanzibar, a semi-autonomous archipelago within the Republic of Tanzania, achieved historic reductions in malaria incidence in the decade following 2002 through extensive interventions [1, 2]. The Zanzibar Malaria Elimination Programme (ZAMEP) sustained these reductions through ongoing case-based surveillance and active follow-up of every malaria case diagnosed across the islands [3, 4]. However, malaria resurgences in 2020 and 2023 have highlighted the fragility of these hard-won gains and the challenges the region faces in pursuing malaria elimination [1, 2, 5, 6]. Key among these challenges is importation of malaria from higher transmission areas, especially mainland Tanzania [1, 7–10]. Zanzibar lies 36.5 km from the coast of Tanzania, with ferries transporting more than 18,000 passengers daily between Dar es Salaam, the main coastal hub, and Unguja, the largest and most populous island in Zanzibar [8, 11]. Thus, like many other areas of sub-Saharan Africa trying to achieve "last mile" malaria elimination, the region remains vulnerable to its re-introduction through human migration. Understanding the contours of human migration, their seasonal variation, and their relationship to malaria cases reported in Zanzibar is crucial for informing targeted interventions [10].

Human mobility data have proven valuable for mapping the distribution and movement patterns of populations vulnerable to malaria [12–14]. Pioneering work used mobile phone and ferry traffic data to measure malaria-relevant human movement flows in Zanzibar, highlighting the importance of returning Zanzibari travellers to malaria importation [8, 9]. Subsequent studies across east and southern Africa have examined within-country and cross-border human population movement [15], supplementing or comparing mobile phone data with national census data [14, 16], travel surveys [17, 18], satellite imagery [19], and genomic data to identify sources and sinks and quantify and map movement flows that drive malaria introduction and transmission [12].

Zanzibar's mobile phone market is now highly fragmented, and privacy concerns have hindered access to cellular data for mobility analysis [20]. Alternative data sources are colocation and movement distribution data, provided by Meta (previously Facebook) Data for Good (DFG). These data have been widely used, especially during the COVID-19 pandemic [21]. To supplement these data, ferry travellers were surveyed at the outgoing Unguja-bound ferry terminal in Dar es Salaam to capture directionality of movement not captured in the Meta data. To examine the seasonality of travel patterns, publicly available rainfall data were obtained to contextualize travel patterns. These data were integrated with comprehensive Zanzibar case data obtained from health facilities in 2022–2023 as part of the larger Zanzibar Imported Malaria (ZIM) project [22].

Overall, this study aims to provide an updated understanding of the role of human mobility and seasonality in shaping travel-associated malaria in Zanzibar. First, factors associated with imported malaria in Unguja are identified. The influence of rainfall patterns in Unguja and mainland Tanzania on human mobility and malaria case incidence in Unguja is then examined. Finally, the Meta colocation data and ferry survey data are used to describe movement patterns between mainland Tanzania and Zanzibar, focusing on differences between mainland and Zanzibari travellers, and comparing the ability of these different data sources to estimate the proportion of malaria importation arising from different mainland regions.

Methods

Within the Zanzibar archipelago, Unguja (6°08′26″S, 39°20′12″E) has historically borne a higher malaria burden, partly due to being the main entry point for travellers. Unguja's climate is characterized by two primary rainy seasons—*masika* (March–May) and *vuli* (October-December)—each followed by a peak in malaria transmission.

Prospective surveys were carried out among malaria cases presenting to health facilities in Unguja and ferry travellers bound for Unguja with approval from the institutional review boards (IRB) in Zanzibar (ZAHREC/01/EXT/July/2023/03), at MUHAS (MUHAS-REC-02–2024-1168), and at UNC (21–1966). Written informed consent, parental consent and/or assent were provided by all participants (Table 1).

Health facility surveys

Surveys were administered to malaria cases diagnosed by rapid diagnostic test (RDT) at health facilities in Unguja from May 2022 to December 2023. A total of 94 health facilities on Unguja were selected using a stratified random sampling approach to draw evenly from urban and

Table 1 Overview of data sources, collection periods, inclusion criteria, and sample sizes

Data Source	Dates of Collection	Inclusion Criteria	Observations (n)	Unit of Observation
Health facility surveys	May 2022 to Dec 2023	RDT-positive individuals classified as a malaria case by facility staff	3,875	Individual person
Ferry surveys	Aug, Sep, Nov, Dec 2023	Ticketed outbound travellers awaiting ferry travel to Zanzibar	1,184	Individual person
Meta movement distribution	Dec 2022 to Aug 2024	iOS or Android users with Facebook app location services enabled	107,683	Daily proportions of users, grouped by distance from home district
Meta colocation	Aug 2022 to Sep 2024	iOS or Android users with Facebook app location services enabled	18,651	Weekly estimates of interaction between districts in Unguja/mainland
CHIRPS rainfall	Dec 2022 to Sep 2024	N/A	N/A	Daily cumulative rainfall (mm) at the district level

rural areas (Suppl. Table 1). Data were collected through standardized Case Report Forms (CRFs), administered by the health facility staff, with training and oversight by ZAMEP malaria surveillance officers. CRFs collected information on demographic factors like age, sex, occupation, and home district, as well as travel history, including (up to two) primary travel destinations, duration of travel, and reason for travel. Cases were classified as travel-associated if they reported travel to mainland Tanzania within the 30 days prior to diagnosis, consistent with national malaria surveillance guidelines in Zanzibar [23]. Survey responses were scanned into an electronic database using ABBYY Flexcapture software (ABBYY Corporation, TX). Data were manually checked and corrected as needed by research staff.

Ferry surveys

Surveys were conducted among ticketed outbound travellers at the Dar es Salaam ferry terminal—including Zanzibaris returning home and mainlanders travelling to Zanzibar—between August and December 2024. Data in this report were drawn from all those screened for survey participation, in which travellers were asked to report their home region or district, whether they were leaving or returning home, travel destination (region or district), number of nights spent away, and number of people travelling with them. Responses were collected manually and entered into Microsoft Excel. Reported regions and districts were matched with those listed in the Database of Global Administrative Boundaries (GADM) [24].

Meta movement distribution data

Movement distribution data from December 2022 to August 2024 were obtained from Meta DFG. Meta DFG data are derived from anonymized location data from users accessing the Facebook mobile app with location services enabled [21]. Movement distribution data provide a profile of the travel originating from each district over time but do not give information on the destinations of travellers. They include a daily proportion of Facebook users with location services enabled who travel a maximum Euclidean distance from their home district within 4 categories: 0 km, 1–9 km, 10–100 km, and > 100 km. To aggregate over a group of districts, the proportions were multiplied by the district population in Tanzania's 2022 census [25] to estimate the number of people from the district in that distance category that day, followed by summation of the estimated counts across districts in each group.

Meta colocation data

Colocation data from mainland Tanzania and Unguja from August 2022 to September 2024 were obtained from Meta DFG. On a weekly basis, Meta determines a likely home district for each individual, and then records how often the locations of pairs of individuals with homes in a given two districts are in the same small geographic area—specifically, the same level 16 Microsoft Bing tile (a spatial unit approximately 600 m by 600 m at the equator). Meta defines the colocation rate as the quotient of co-observation (probability that two individuals are observed simultaneously) and colocation (probability that two individuals are observed simultaneously in the same location) [21]. These estimates are provided weekly for each pair of Tanzania's 170 administrative districts (median population 306,169, range 63,156–1,649,912) and represent the degree of interaction between them. The data were first cleaned by removing district-district pairs with three or fewer weekly observations. Observations where either the measured co-observation or measured colocation rates were outliers (> 1.5*IQR outside of IQR) for that district-district pair were then dropped. For each Zanzibar-mainland district pair, the average over time of the observed colocation and observed co-observation was calculated, then their quotient as the district-district colocation rate was calculated. The

district-district colocation rate was multiplied by both districts' population to give weighted colocation. Then, the weighted colocation was summed over all districts in Unguja and districts within a mainland region to obtain an estimate for the total interaction between all of Unguja and each mainland region. Meta does not release data on the counts behind the proportions, so there is no known sample size, other than the privacy floor of 10 people per district per day.

Statistical analysis

From the health facility surveys, the most frequently reported destinations were determined for those who reported travel to the mainland in the previous 30 days. Odds ratios (ORs) and 95% confidence intervals (CIs) were calculated to compare the odds of being a travel-associated case versus a locally acquired malaria case across demographic characteristics. Adjusted ORs (aORs) were estimated using a mixed effects logistic regression, accounting for age, sex, occupation, seasonality of malaria infection, and random effects of the case's home district. Factors associated with short- and long-term travel (defined as 14 or fewer versus greater than 14 nights away) were compared using aORs and 95% CIs, controlling for random effects of the case's home district.

Using Meta movement distribution data, the proportions of travellers travelling long (> 100 km) and medium (10–100 km) distances on rainy vs. dry days (cumulative rainfall over the previous 30 days of greater or less than 100 mm, respectively) were compared using Mann–Whitney U-testing. Data on regional rainfall were obtained from Climate Hazards Group InfraRed Precipitation with Station (CHIRPS) data [26] through Google Earth Engine [27]. The relationship between weekly malaria case counts for travellers from each travel destination region and rainfall at that region was modelled using multivariate negative binomial regression. Lagged cumulative rainfall in both Zanzibar and the relevant mainland region were included as covariates in each model. Cumulative rainfall from 2 to 6 weeks prior to clinic visit date was calculated to account for vector ecology, the incubation period of malaria, and median travel time [28]. A full model was fit for all travellers, and separate models were fit for travel subgroups, including travellers spending more than three nights away and those visiting specific mainland regions. An additional model was fit for non-travellers residing in Zanzibar. When assessing associations with seasonality of malaria infection, the long rainy season and short rainy season were defined as March–May and October-December, respectively [29].

Finally, proportions of travellers arriving from common travel destinations were compared based on data derived from Meta colocation, ferry screening, and Zanzibar malaria clinic cases. Results were visualized in proportion plots which compared relative proportions of travellers captured by each data source. Region- and district-level malaria risk strata were obtained from Tanzania's 2021–2025 National Malaria Strategic Plan [6] (Suppl. Figure 1). For regional analyses, Kagera, Mwanza, Geita, Shinyanga, Simiyu, and Mara were grouped as the "Lake Zone" [30]. For all analyses, significance was tested at an α level of 0.05, with *p*-values adjusted for multiple testing using the Benjamini–Hochberg procedure. Analyses were performed using R (4.4.2), Python (3.10.14), and Tableau Public (2024.3.0).

Results

Factors associated with travel-related malaria

Among 3,875 individuals who presented to health facilities in Unguja with malaria from May 2022 to December 2023, 1,172 (30%) reported travel to mainland Tanzania within the past 30 days and were presumptively categorized as imported or travel-associated malaria cases. Travel-associated cases were dominant in 2022, composing 55% (283/511) of captured malaria cases, while only 27% (889/3,348) of cases in 2023, which saw a substantial increase in cases, were classified as travel-associated. The majority of imported cases were reported in central Unguja, including the urban districts of Mjini and Magharibi. Travellers most often visited Tanga (n = 238, 20%), Dar es Salaam (n = 231, 20%), Morogoro (n = 170, 15%), and Pwani (n = 137, 12%) regions; 144 cases (12%) reported travelling to the Lake Zone (Fig. 1).

Children were more likely to have travel-associated malaria, with malaria cases under five years of age having 3.2 times the odds of reporting recent travel to the mainland (95% CI: 2.2–4.8). Females with malaria were also more likely to report recent travel to the mainland (aOR = 1.5, 95% CI: 1.3–1.8) (Table 2). In terms of occupation, farmers were more likely to have travel-associated malaria (aOR = 1.6, 95% CI: 1.2–2.1) compared to those engaged in trade or business. In contrast, night watchmen (aOR = 0.3, 95% CI: 0.2–0.4) and students (aOR = 0.5, 95% CI: 0.3–0.7) were less likely to report travel to the mainland and conversely more likely to have locally acquired malaria, instead reporting travel within Zanzibar, along with those working in fishing, tourism, or public service/NGOs (Table 3). Finally, malaria cases presenting during the dry season had 1.5 times the odds of being associated with travel to the mainland (95% CI: 1.3–1.8).

More than half of malaria cases who traveled to the mainland (52%, 520/1,001) spent more than 14 nights away from home (Suppl. Figure 2). These long-term travellers were more likely to have visited mainland regions

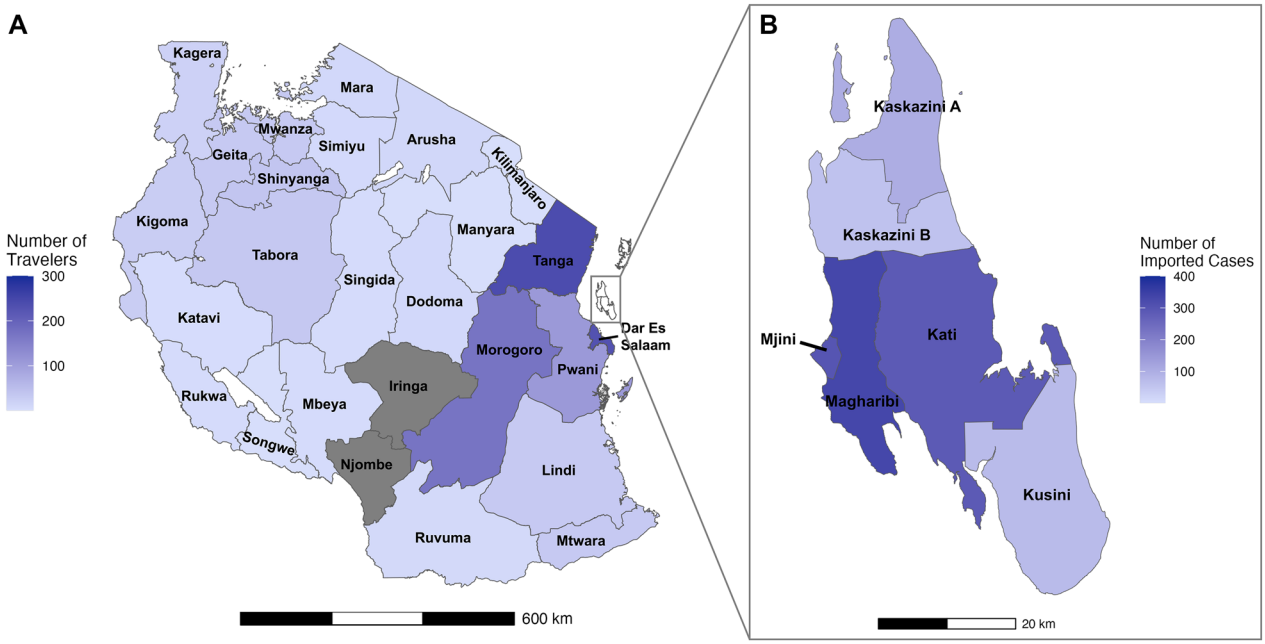

Fig. 1 (**A**) Self-reported travel destinations for confirmed malaria cases on Zanzibar who reported recent travel to the mainland and (**B**) home districts of travel-associated malaria cases on Zanzibar, May 2022 to December 2023 (n = 1,172). Darker blue indicates more travellers, while lighter blue represents fewer travellers. Mainland regions shown in dark grey indicate no recorded travel from Zanzibar

considered to be at high or moderate risk for malaria (aOR = 2.1, 95% CI: 1.5–3.0; aOR = 1.6, 95% CI: 1.2–2.2, respectively) (Table 4). Visiting family was the leading reason for travel among both long- and short-term travellers (61% (302/520) and 57% (259/481), respectively). Those visiting family had slightly increased odds of travelling for extended durations (aOR 1.4, 95% CI: 1.1–2.0), compared to those travelling for work.

A sizable proportion of Zanzibar malaria cases were found in those with strong ties to the mainland. Among all Unguja malaria cases surveyed, 15% (600/3,875) of all cases and 28% (333/1,172) of travel-associated cases reported their primary residence to be on the mainland. Almost two-thirds (372/600, 62%) of mainland cases reported having resided less than 6 months on Zanzibar; 56% (333/600) reported visiting the mainland within the past month (Suppl. Table 2). Compared to Zanzibaris with malaria, mainland cases were more likely to report farming as their primary occupation (22% of mainlanders vs. 15% of Zanzibaris, $p < 0.001$), especially among those who had resided in Zanzibar for less than 6 months (24%) or recently traveled to the mainland (28%).

Rainfall associations with mobility and malaria in Zanzibar

Imported malaria cases appeared to peak outside of the rainy season months in Zanzibar (March–May and October-December) during the study years (Suppl. Figure 3). One reason for this may be a negative effect of rainfall on human mobility that drives travel-related malaria exposure. Based on movement distribution data from December 2022 to August 2024 totaling 107,683 daily observations, 3–7% of Zanzibari Facebook users traveled medium distances (10–100 km) on a monthly basis (Fig. 2A). This distance includes movement between districts within Zanzibar or to nearby coastal cities on the mainland, such as Dar es Salaam. An even smaller fraction, only 0.2–1.2% of Facebook users in Zanzibar, traveled long distances (more than 100 km) on a monthly basis (Fig. 2C). Both medium- and long-distance travel decreased during the long rainy season (March–May), especially from urban districts, where travel generally surpassed that in rural districts. Part of the long rainy season also overlapped with Ramadan in 2023 (23 March–20 April) and 2024 (10 March–9 April), which may have additionally reduced travel. Less pronounced dips in both types of travel were also observed during the short rainy season (October-December). Overall, both long- and medium-distance travel were lower on days when the previous 30 days of accumulated rainfall surpassed 100 mm ($p < 0.001$ for both comparisons).

Another reason that imported malaria cases in Zanzibar may peak outside the rainy season is the rainfall patterns of mainland Tanzania differ from that in Zanzibar. Imported malaria case counts were associated with rainfall in the mainland regions visited, with a 6% increase in cases per inch increase in rainfall (IRR: 1.06, 95% CI:

Table 2 Odds of travel-associated malaria compared to locally-acquired malaria among Zanzibari malaria cases, 2022–2023

	Total (N = 3875)	Travellers (N = 1172)	Non-travellers (N = 2703)	OR (95% CI)	aOR (95% CI)
Sex					
Male	2604 (67.2%)	679 (57.9%)	1925 (71.2%)	REF	REF
Female	1265 (32.6%)	489 (41.7%)	776 (28.7%)	1.8 (1.5–2.1)	1.5 (1.3–1.8)
Age					
< 5	238 (6.1%)	130 (11.1%)	108 (4.0%)	3.1 (2.4–4.1)	3.2 (2.2–4.8)
5–15	496 (12.8%)	170 (14.5%)	326 (12.1%)	1.4 (1.1–1.7)	1.5 (1.0–2.2)
> 15	3141 (81.1%)	872 (74.4%)	2269 (83.9%)	REF	REF
Occupation					
Trader/business	762 (25.4%)	219 (25.8%)	542 (25.2%)	REF	REF
Farming	416 (13.9%)	192 (22.6%)	224 (10.4%)	2.1 (1.7–2.7)	1.6 (1.2–3.1)
Housewife	341 (11.4%)	126 (14.8%)	215 (10.0%)	1.5 (1.1–1.9)	–
Watchman/security	233 (7.8%)	20 (2.4%)	213 (9.9%)	0.2 (0.1–0.4)	0.3 (0.2–0.4)
Student	211 (7.0%)	35 (4.1%)	176 (8.2%)	0.5 (0.3–0.7)	0.5 (0.3–0.7)
Fishing	182 (6.1%)	44 (5.2%)	138 (6.4%)	0.8 (0.5–1.1)	0.8 (0.5–1.2)
Public servant/NGO	131 (4.4%)	34 (4.0%)	97 (4.5%)	0.9 (0.6–1.3)	0.8 (0.5–1.2)
Tourism	96 (3.2%)	31 (3.7%)	65 (3.0%)	1.2 (0.7–1.9)	1.2 (0.8–2.0)
Food service	95 (3.2%)	25 (2.9%)	70 (3.3%)	0.9 (0.5–1.4)	0.8 (0.5–1.2)
Construction	48 (1.6%)	14 (1.6%)	34 (1.6%)	1.0 (0.5–1.9)	1.1 (0.5–2.1)
Other	487 (16.2%)	109 (12.8%)	378 (17.6%)	–	–
Missing	43	13	30	–	–
Season					
Rainy	1940 (50.1%)	492 (42.0%)	1448 (53.6%)	REF	REF
Dry	1935 (49.9%)	680 (58.0%)	1255 (46.4%)	1.6 (1.4–1.8)	1.5 (1.3–1.8)

Odds ratios (ORs) and 95% confidence intervals (CIs) are presented for each demographic characteristic. Adjusted ORs were estimated using a mixed effects logistic regression, controlling for age, sex, occupation, seasonality, and district-level random effects. A ratio greater than 1 indicates higher odds of a case being travel-associated. Season refers to the period during which the patient presented to the health facility.

1.04–1.08). Meanwhile, non- travel-associated malaria cases on Zanzibar were associated with recent rainfall in Zanzibar, with a 10% increase in cases per inch increase in rainfall (IRR: 1.10, 95% CI: 1.05–1.15) (Fig. 3).

Mainland-Zanzibar travel patterns

Meta colocation data and ferry traveller survey data differed in their depiction of human movement between mainland Tanzania and Zanzibar (Fig. 4). Colocation data, which depict population mixing among mobile phone owners with Facebook location services turned on, measured 18,651 weekly colocation estimates between a district in Unguja and another district in Tanzania from December 2022 to September 2024. Regions along the eastern coast and northeast border of Tanzania, including Dar es Salaam, Tanga, Kilimanjaro, Pwani, and Arusha, were the top mainland regions exhibiting interactions with Zanzibaris (Fig. 4A). In terms of interactions with individual districts within Unguja, even though the main ferry terminal on the mainland is in Dar es Salaam, the top-ranking mainland-Unguja colocation district pairs were Pangani-Mjini, Mwanga-Kati, and Pangani-Kaskazini A (Suppl. Figure 4). Mwanga, in Kilimanjaro, is in a low malaria risk region. However, Pangani is a fishing port in Tanga—a moderate malaria risk region—where boats can travel to Zanzibar City (Mjini) after stopping in Pemba, as well as to official and unofficial ports in Kaskazini A. Colocation interactions did not differ in rainy compared to dry season months (Suppl. Figure 5).

Unlike colocation data, ferry survey data provide directionality of travel. Among 1,184 travellers screened at the Unguja-bound ferry terminal in Dar es Salaam between August–September and November–December 2024, two-thirds were mainlanders travelling to Zanzibar (n = 795, 67%) while one-third were Zanzibaris returning home (n = 389, 33%). Only 26% (103/389) of returning Zanzibari ferry travellers reported spending more than 14 nights away (Suppl. Figure 2), in contrast to Zanzibari clinic cases, of whom a majority did so (52%, 520/1001). Top mainland origins or travel destinations, weighted by duration of travel capped at 28 days, were Dar es Salaam (7189, 43%), Pwani (1752, 10%), Morogoro (1437, 8.6%), Tanga (850, 5.1%), and Dodoma (579, 3.5%), where the

Table 3 Select demographic characteristics of confirmed malaria cases on Zanzibar who did not report travel to the mainland within the past 28 days, 2022–2023

	Total (N = 2582)	Travel within Zanzibar (N = 473)	No travel (N = 2109)	OR (95% CI)	aOR (95% CI)
Sex					
Male	1835 (71.1%)	347 (73.4%)	1488 (70.6%)	REF	REF
Female	745 (28.9%)	126 (26.6%)	619 (29.4%)	0.9 (0.7–1.1)	1.1 (0.9–1.4)
Age					
< 5	97 (3.8%)	13 (2.7%)	84 (4.0%)	0.7 (0.4–1.2)	0.8 (0.4–1.6)
5–15	308 (11.9%)	51 (10.8%)	257 (12.2%)	0.9 (0.6–1.2)	1.1 (0.6–1.9)
> 15	2177 (84.3%)	409 (86.5%)	1768 (83.8%)	REF	REF
Occupation					
Trader/business	522 (25.3%)	72 (18.3%)	450 (26.9%)	REF	REF
Farming	213 (10.3%)	41 (10.4%)	172 (10.3%)	1.5 (1.0–2.3)	1.4 (0.9–2.2)
Housewife	209 (10.1%)	24 (6.1%)	185 (11.1%)	0.8 (0.5–1.3)	–
Watchman/security	205 (9.9%)	59 (15.0%)	146 (8.7%)	2.5 (1.7–3.7)	2.6 (1.7–3.8)
Student	171 (8.3%)	42 (10.7%)	129 (7.7%)	2.0 (1.3–3.1)	2.0 (1.3–3.1)
Fishing	130 (6.3%)	31 (7.9%)	99 (5.9%)	2.0 (1.2–3.1)	1.9 (1.2–3.1)
Public servant/NGO	94 (4.5%)	24 (6.1%)	70 (4.2%)	2.1 (1.3–3.6)	2.1 (1.3–3.6)
Food service	67 (3.2%)	13 (3.3%)	54 (3.2%)	1.5 (0.8–2.9)	1.5 (0.8–2.8)
Tourism	63 (3.0%)	19 (4.8%)	44 (2.6%)	2.7 (1.5–4.9)	2.7 (1.5–4.9)
Construction	33 (1.6%)	4 (1.0%)	29 (1.7%)	0.9 (0.3–2.5)	0.9 (0.3–2.6)
Other	359 (17.4%)	64 (16.3%)	295 (17.6%)	–	–
Missing	28	4	24	–	–
Season					
Rainy	1381 (53.5%)	251 (53.1%)	1130 (53.6%)	REF	REF
Dry	1201 (46.5%)	222 (46.9%)	979 (46.4%)	1.0 (0.8–1.2)	1.0 (0.8–1.3)

Odds ratios (ORs) and 95% confidence intervals (CIs) are presented for each demographic characteristic. Adjusted ORs were estimated using a generalized linear mixed-effects model, controlling for age, sex, occupation, seasonality, and district-level random effects. A ratio greater than 1 indicates higher odds of a case travelling within Zanzibar

Table 4 Malaria risk of region visited and self-reported reason for travel for Zanzibar confirmed malaria cases who reported recent travel to the mainland, 2022–2023

	Total (N = 1001)	Short-term (≤ 14 nights) (N = 481)	Long-term (> 14 nights) (N = 520)	aOR (95% CI)
Malaria risk of region visited				
Very low/low	290 (29.7%)	168 (36.2%)	122 (23.8%)	REF
Moderate	408 (41.8%)	188 (40.5%)	220 (43.0%)	1.6 (1.2–2.2)
High	278 (28.5%)	108 (23.3%)	170 (33.2%)	2.1 (1.5–3.0)
Missing	25	17	8	–
Reason for travel				
Work	228 (24.1%)	126 (27.8%)	102 (20.7%)	REF
Visiting family	561 (59.3%)	259 (57.0%)	302 (61.4%)	1.4 (1.1–2.0)
Vacation	31 (3.3%)	18 (4.0%)	13 (2.6%)	0.9 (0.4–1.9)
Other	126 (13.3%)	51 (11.2%)	75 (15.2%)	
Missing	55	27	28	

Adjusted odds ratios (aORs) and 95% confidence intervals (CIs) were calculated to assess the relationship between travel duration and malaria prevalence, as well as between travel duration and reason for travel. Analyses controlled for cases' home districts as a random effect. Malaria risk estimates are derived from the 2021–2025 Tanzania National Malaria Strategic Plan. Travellers with missing travel duration data were excluded from this table

total person-travel days among all travellers was 16,769 (Fig. 4B). A greater proportion of mainlanders versus Zanzibaris were travelling from Morogoro, Tanga, Mtwara, and Geita, while a greater proportion of Zanzibaris were returning from Pwani and Dodoma (Suppl. Figure 6). Overall, the colocation data highlight population mixing in coastal and northern regions, whereas ferry surveys show travel arriving to Zanzibar from coastal as well as more interior regions (Fig. 4C).

Linking human mobility to malaria importation

Examining how well the colocation and ferry data predict the origin of malaria cases in Unguja highlights the role of malaria endemicity and modes of travel. Travel to mainland regions with the lowest malaria endemicity (Kilimanjaro, Arusha) does not translate to malaria cases while travel to regions with high malaria risk (Lake Zone) is overrepresented in the clinic cases, relative to the colocation-derived population mixing, and to a lesser degree, importation on the ferry (Fig. 4D). The ferry data better reflect travel-related malaria from the coastal Pwani and neighboring Morogoro region. However, travel from Dar es Salaam is substantially overrepresented in the ferry data compared to the proportion of clinic cases reporting travel from Dar es Salaam (45% ferry vs. 22% clinic). Also,

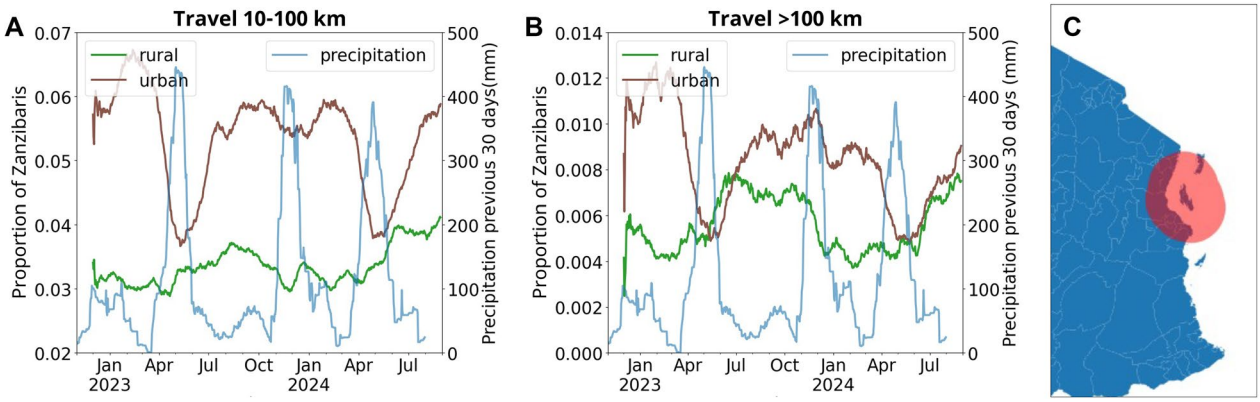

Fig. 2 Proportion of Zanzibaris travelling (**A**) medium distances (10–100 km) or (**B**) long distances (> 100 km) from their home district (classified as rural or urban), plotted alongside cumulative precipitation over the previous 30 days, 2023–2024. Urban districts include Mjini and Magharibi; rural districts include Kaskazini A & B, Kusini, and Kati. Left y-axes show the proportion of Zanzibaris travelling while the right y-axes show precipitation in mm; the left axes have differing scales, due to the larger proportion of medium-distance travel. Travel estimates are from Meta movement distribution data, based on anonymized location data from Facebook users in Unguja with location services enabled. (**C**) Map showing a 100 km radius around Unguja (shaded in red), encompassing common medium-distance destinations such as Dar es Salaam and coastal parts of Tanga and Pwani regions

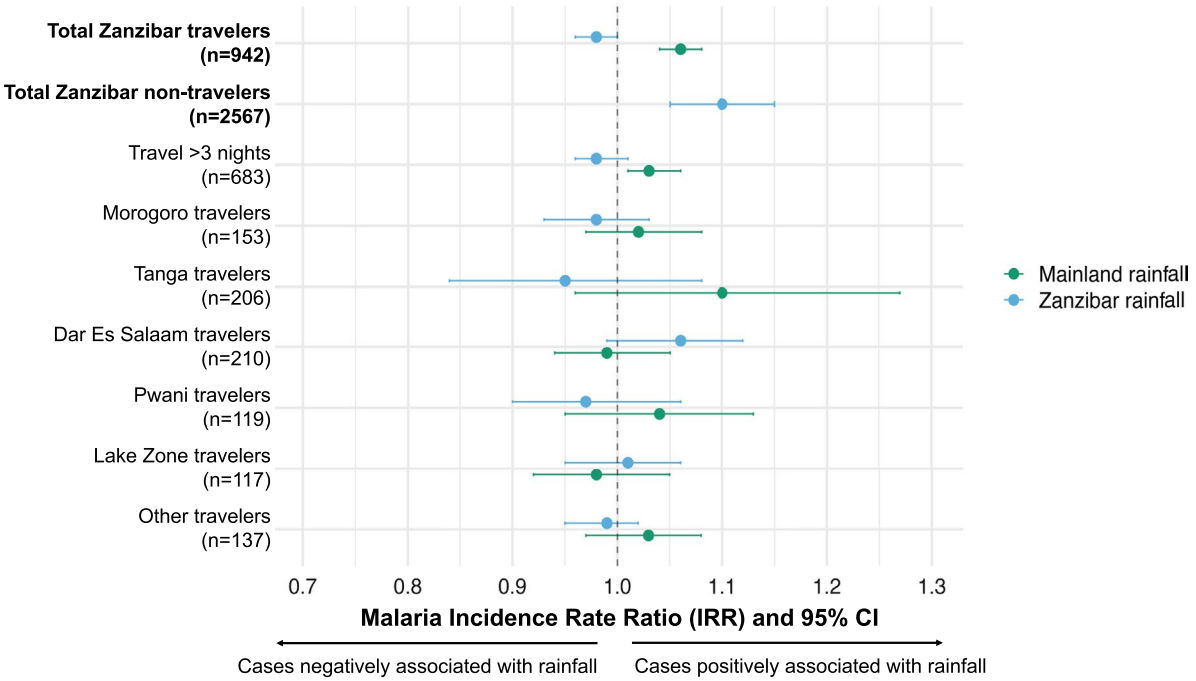

Fig. 3 Association between malaria cases in Zanzibar and regional rainfall. The figure presents the results of negative binomial regression models evaluating the association between malaria case counts in Zanzibar and cumulative regional rainfall (Zanzibar and mainland Tanzania) for cases occurring from October 2022 onward. A 2-week lag prior to clinic visit was applied for each case, and rainfall was summed over the preceding 4 weeks. Malaria cases were grouped by the travel region visited, and the models assessed the relationship between case counts and rainfall in the corresponding region. Each point depicts the incidence rate ratio (IRR) with its 95% confidence interval (CI). Associations with mainland rainfall are shown in green, and those with Zanzibar rainfall are in blue. Results are shown for all Zanzibar travellers and non-travellers, as well as for key subgroups: travellers spending more than 3 nights at their destination and those visiting common destinations

at-risk travel to Tanga appears to be underestimated by the ferry data (5% ferry vs. 22% clinic). The overall colocation data better depict at-risk travel from Tanga; however, imported cases in the northern part of Unguja (Kaskazini A&B districts) associated with travel to Tanga are still substantially underestimated in the colocation data

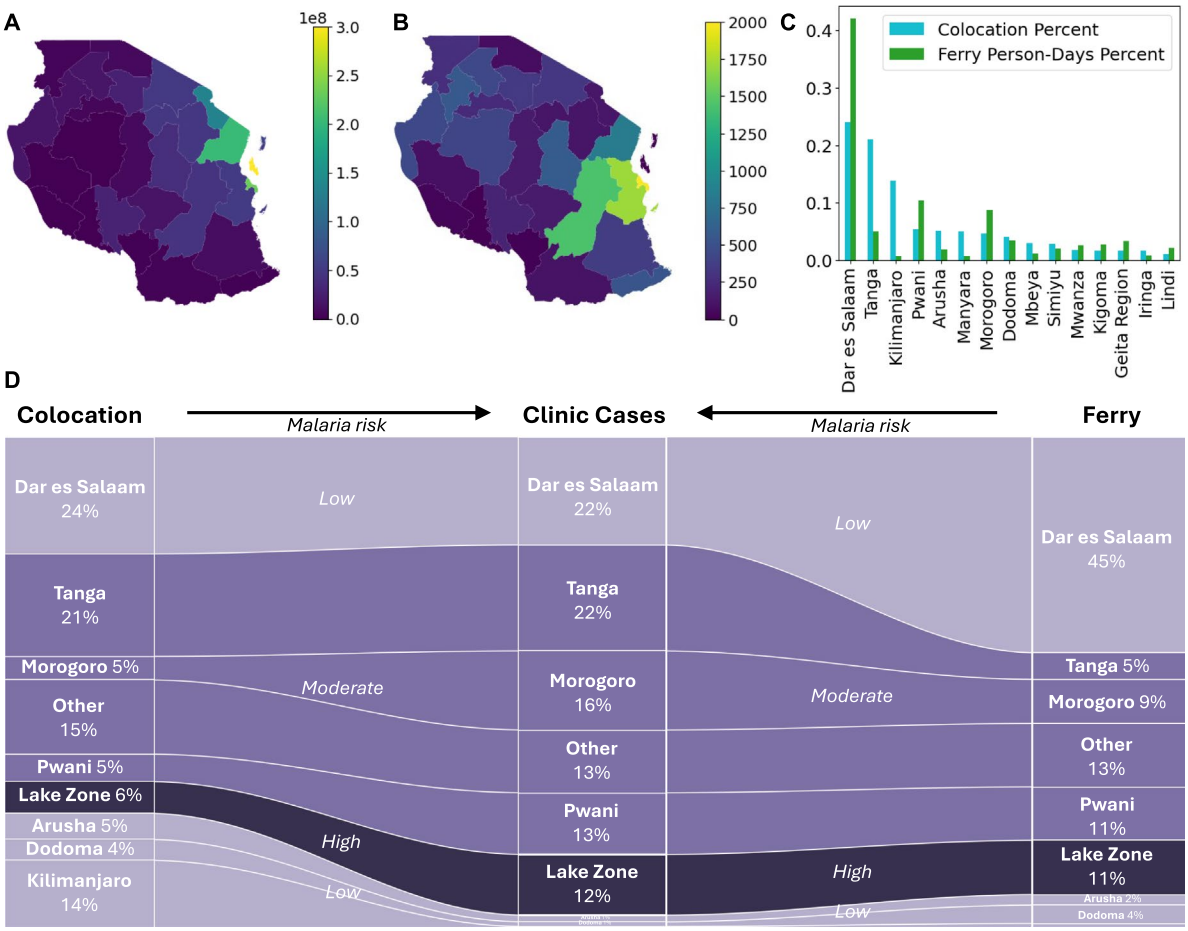

Fig. 4 Estimation of human mobility between Unguja and mainland Tanzania. (**A**) Weighted colocation between Unguja and mainland Tanzania, aggregated between December 2022 and September 2024. Colocation represents the probability that individuals from these regions are within a small distance of one another, weighted by population. (**B**) Origins and travel destinations of ferry passengers travelling from Dar es Salaam to Unguja in August–September and November–December 2024, weighted by the length of travel (up to 28 days). (**C**) Comparison of the percent contribution of regions to travel between the mainland and Unguja, according to both the colocation data and ferry survey data. (**D**) Relative proportions of Zanzibar travellers captured through colocation data (left) and ferry surveys (right), compared to travel-associated malaria clinic cases (middle). Colours represent regional malaria risk, with darker shades indicating higher risk. Depicted risk is an average of region-level malaria risk strata (from the 2021 to 2025 Tanzania National Malaria Strategic Plan), weighted by population. Note that travel to Kilimanjaro is not represented for clinic cases or ferry travellers, as it accounts for less than 1% of travel volume

(Suppl. Figure 7). Compared to mainlanders on the ferry, there were fewer Zanzibaris visiting the Lake Zone and Morogoro, but more of this travel appeared to translate into imported malaria cases from these regions (Suppl. Figure 8).

Overall, taking into account travel duration, primary residence (mainland vs. Zanzibar), seasonality of travel (wet vs. dry), and malaria risk of travel destination (moderate/high vs. low), the highest proportion of travel-related malaria cases was seen in Zanzibari travellers, both long- and short-term, who reported travel to mainland districts with high/moderate malaria risk in the dry season (Fig. 5). Among surveyed ferry travellers, only 2.1% (19/899) fell into this high-risk category, despite accounting for 32% (309/977) of imported malaria cases.

Discussion

Rapid demographic growth, urbanization, and strong interconnectivity with mainland Tanzania are likely to increase human movement between Zanzibar and mainland Tanzania over time, keeping Zanzibar susceptible to malaria importation [11]. This study describes imported malaria cases to Unguja in 2022–2023 and links these case data to trends in human mobility and rainfall seasonality to understand the drivers of malaria importation that can inform future intervention. These data suggest that a few mainland regions disproportionately

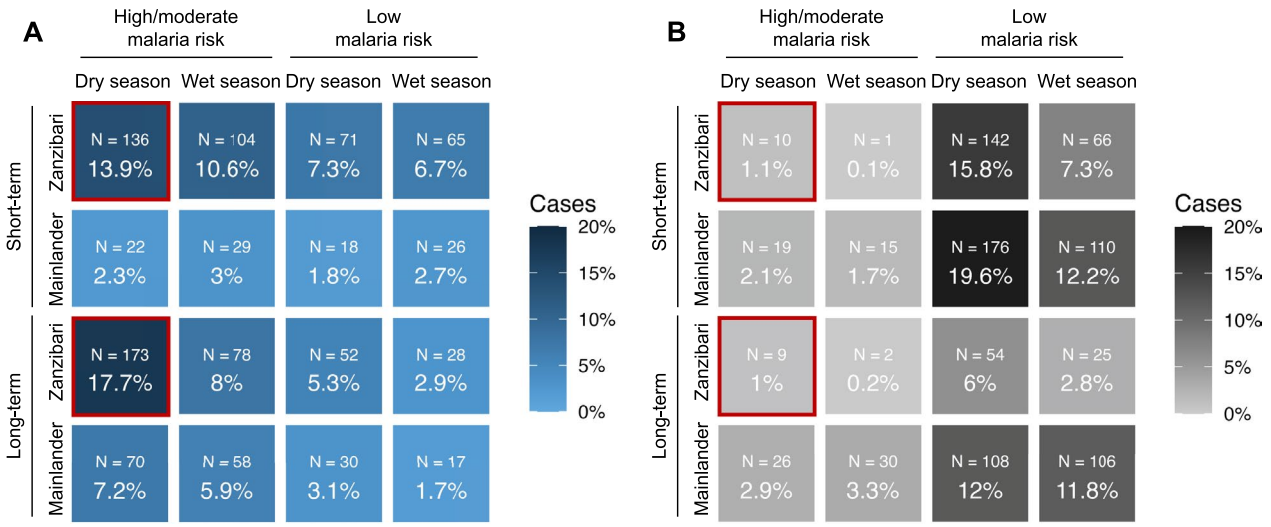

Fig. 5 Proportions of travellers by key demographic and travel-related factors. The heat maps display the proportions of (**A**) Zanzibari malaria cases who reported recent travel to the mainland and (**B**) travellers at the outbound Dar es Salaam ferry terminal. Both heat maps are categorized by four factors: malaria transmission risk of the mainland district visited, Zanzibar season during travel (wet/dry), duration of travel (short/long; long defined as greater than 14 nights at the destination), and self-reported home residence (Mainlander vs. Zanzibari). Each cell represents the percentage of total travellers corresponding to specific combinations of these factors. Target populations of interest are marked in red

contribute to imported malaria to Zanzibar due to their higher malaria endemicity and connectivity to Unguja—in particular, Tanga, Pwani, Morogoro, and the Lake Zone. Importantly, human mobility data, especially those from the Meta DFG platform, approximate the composition of at-risk travel relatively well. As might be expected, travelling for longer duration poses greater risk; children and females/housewives, likely visiting family in these regions, were more likely to have travel-associated malaria, as were farmers, who made up the largest group of mainlanders with recent travel to the mainland.

Zanzibari residents make up the bulk of imported malaria cases, but young men migrating from the mainland to work and farm in Zanzibar who maintain ties and travel back to the mainland are also a key risk population. Mobile and migrant populations present challenges to malaria control due to their diverse movement patterns and heightened exposure to malaria vectors, often coupled with limited access to interventions [31–33]. Malaria outbreaks in urban districts in Unguja in 2023 were linked to night watchmen/police, construction workers, and farmers, occupations commonly reported by surveyed mainlanders (Suppl. Table 2) [23, 34]. While research conducted 15 years ago estimated that Zanzibari residents travelling to malaria endemic regions contributed up to 15 times more imported cases than infected mainland visitors [9], this pattern may be shifting as more mainland residents travel to Zanzibar over time [11]. It is also plausible that mainlanders—who travel to regions with higher malaria risk for longer periods of time (Suppl. Figure 6)—are more likely to be semi-immune, leading them to seek care less frequently but still contribute to local transmission via chronic asymptomatic carriage [35, 36].

The seasonality of at-risk travel is likely complex. Meta movement distribution data showed a negative association between local rainfall and travel, especially from urban districts where many travellers reside. However, colocation data did not show seasonal shifts in the patterns of interaction between mainland regions and Unguja. Disparate rainfall patterns at source or destination regions likely drive exposure to malaria and the timing of imported cases. These findings may help explain previously observed patterns, including increased malaria risk associated with travel during the dry season (July–September) [10], and a 12–13 week lag in cross-correlation analysis of weekly precipitation on Zanzibar and weekly reported index cases [3].

These findings support a strategy targeting high-risk travellers for preventive interventions to reduce malaria importation [37–41]. In a 2017–18 study using reactive case detection data, travellers to regions of high malaria endemicity contributed to 77% of all travel-related infections. These health facility data similarly highlight the outsized contribution of travel to more endemic areas. Although mainlanders outnumbered Zanzibari travellers 2.2 to 1 in this survey of ferry travellers, Zanzibari travellers account for more imported malaria cases (2.6 to 1) (Suppl. Figure 7). Targeting Zanzibari travellers travelling to moderate- or high-risk malaria regions in the dry

season months could mean intervening on less than 5% of ferry travellers while potentially preventing up to one-third of imported malaria cases (Fig. 5). This could be presumptive treatment or prophylaxis at ports of departure or entry [32, 42].

Prior modelling work suggests that border post interventions are most effective in near-elimination settings like Zanzibar, especially when targeting major travel entry points [43]. This study highlights Dar es Salaam as a primary travel hub and destination linking Unguja and mainland Tanzania. However, substantial travel and malaria importation from Tanga were also observed, likely arriving via vessels to official and unofficial ports in Kaskazini A [11]. Thus, digital mobility data can highlight populations underrepresented using traditional survey methods that would need to be considered in any proposed intervention [9, 43].

This study has some key limitations. First, imported cases were categorized based on travel within the previous 30 days, but residual local transmission may have led to some of these cases being acquired locally, especially if travel was limited to Dar es Salaam. Conversely, more remote travel that was not elicited could in some cases lead to chronic parasite carriage and/or persistence of RDT-positivity that is not classified as travel-associated malaria. Second, there was no control group of travellers without malaria to directly assess travel factors associated with malaria infection. Instead, ferry travel and digital mobility data were leveraged to compare overall travel patterns to those of travel-associated malaria cases. Reassuringly, Fakih et al., using reactive case detection data to compare qPCR-positive vs. negative travellers, generated findings that mirror these—specifically, that longer duration of travel to highly malaria-endemic regions increased risk of infection [10]. Third, bias introduced by use of Meta DFG data could not be quantified [44]. Meta does not release numbers of users; all data are provided as proportions. Thus, it is not possible to speculate about the representativeness of the sample to the general Zanzibari population or attempt to correct for potential biases. While these data likely reflect broader population movement patterns, they do not directly represent confirmed malaria cases or infection status. However, social media uptake is highest in young adults, the major malaria risk group in Zanzibar [20]. Additionally, global migration flows estimated using Meta data have been validated against high-quality administrative datasets [45]. Despite potential for bias, colocation data approximated the proportional makeup of at-risk travel well and provided more fine-scale spatial data compared to mobile phone coverage, which can comprise a variety of malaria risk regions [8]. Finally, the movement data, case data, and ferry survey data were collected at different times. Future work that aligns survey periods across data sources would help strengthen inferences.

In summary, malaria importation to Zanzibar is driven by high-risk travel from a few mainland regions, influenced by seasonal and occupational mobility. Targeted interventions informed by both mobility data and identified risk groups may offer an effective strategy for reducing importation.

Abbreviations

CHIRPS	Climate Hazards Group Infrared Precipitation with Station data
CRF	Case report form
DFG	Meta Data for Good
GADM	Database of Global Administrative Boundaries
IRB	Institutional review board
IRR	Incidence rate ratio
MUHAS	Muhimbili University of Health and Allied Sciences
RDT	Rapid diagnostic test
UNC	University of North Carolina
ZAHREC	Zanzibar Health Research Ethics Committee
ZAMEP	Zanzibar malaria elimination program
ZIM	Zanzibar Imported Malaria

Supplementary Information

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Additional file 1

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Author contributions

Conception and design: JTL, JJJ, VG, BEN, RR, AB, ME, DB; Data acquisition, analysis, and interpretation: JGM, KD, VG, MA, WH, BM, BBC, DM, AZ, MN, RK, SJS, JTL; Drafting manuscript: JGM, KDM, JTL, RR; Revision and final approval: All authors; Accountability: JGM, KD, BEN, JJJ, JTL.

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Availability of data and materials

The clinic datasets analysed during this study are available in the UNC Dataverse repository, https://doi.org/10.15139/S3/F6FKRL. Colocation and movement distribution data were obtained from Meta Data for Good and are available subject to Meta's Data Use Agreement.

Declarations

Ethics approval and consent to participate

Ethical approval was obtained from the institutional review boards (IRB) in Zanzibar (ZAHREC/01/EXT/July/2023/03), at MUHAS (MUHAS-REC-02-2024-1168), and at UNC (21-1966). Written informed consent, parental consent, and/or assent were provided by all participants.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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